RESQ-AI // COMMAND CENTER GUIDE

Live Dashboard URL: https://sih-2026-five-sable.vercel.app/

SIH 2026
Autonomous Search & Rescue

Team Handout & Architecture

1. Executive Summary & Project Purpose

RESQ-AI is an end-to-end autonomous search-and-rescue platform engineered for disaster response organizations (such as NDRF, SDRF, and military triage teams). When floods, earthquakes, or avalanches strike, manual search is dangerously slow. RESQ-AI combines autonomous drone navigation, high-frequency 5Hz telemetry, real-time edge computer vision (YOLOv8), acoustic alerting, and immediate digital handover to ground rescue units.

2. How the System Works (The 4 Core Components)

Component	Technology	Operational Role
1. Frontend Dashboard	Next.js 14, Mapbox GL, Tailwind, Web Audio	Tactical HUD showing 3D maps, live video stream, avionics deck, survivor triage queue, and mission controls.
2. API Gateway	Node.js, Express, Socket.IO	Central hub routing 5Hz telemetry, raw JPEG binary frames, mission targets, and detection alerts with zero latency.
3. Flight Worker	Python, Kinematics / DroneKit SITL	Autopilot simulator managing vertical climb (30m), waypoint heading, horizontal travel, and battery drain.
4. AI Vision Worker	Python, Ultralytics YOLOv8, OpenCV	Runs high-speed neural network inference on drone aerial camera frames, tags human survivors, and posts alerts.

3. Interactive Dashboard Tour

- **Tactical 3D Mapbox View**: Displays live drone position (blue pulse), destination targets (yellow), and confirmed survivor locations (red pins). Restricted by a built-in 3.0 km safety geofence.
- **Avionics & Telemetry Deck**: Real-time indicators showing altitude (30.0m AGL), battery discharge percentage, and 5-decimal precision GPS coordinates.
- **AI Gimbal Vision HUD**: Live aerial camera feed with targeting crosshairs, camera field-of-view (84° FOV), and bounding box overlays identifying survivors (e.g., 'PERSON 92.4%').
- **FLIR Thermal Vision Simulation**: One-click filter switching the video feed into infrared white-hot contrast for nighttime disaster operations.
- **Acoustic Alert Synthesizer**: Synthesizes dual-tone tactical alert beeps (880 Hz / 1320 Hz) via Web Audio API whenever a survivor is confirmed.
- **Survivor Triage Queue**: Chronological feed of all detections. Clicking 'Export Manifest' downloads a structured JSON/CSV report for field rescue teams.
- **RTL (Return to Launch)**: Autonomous safety recall button that commands the drone to navigate straight back to the home base (12.9716°N, 77.5946°E).

4. How Team Members Can Test the System

Step 1: Open https://sih-2026-five-sable.vercel.app/ on any desktop or mobile browser.

Step 2: Click the **'TEST AUDIO'** button in the top bar to verify sound output on your device.

Step 3: Toggle **'FLIR THERMAL'** to preview night-vision infrared simulation mode.

Step 4: On the demonstration laptop, launch **start_demo.bat** to spin up the API and AI workers.

Step 5: Click **'QUICK DEMO MISSION'** or click anywhere on the tactical map to deploy a search route.

Step 6: Watch the live flight progress, hear the alarm chime, and click **'EXPORT MANIFEST'** to inspect the digital rescue handover file.

5. The 30-Second Winning Pitch Script for Judges

> *"Judges, in disaster management, the first 72 hours are critical. Manual search across large debris fields is slow and perilous. RESQ-AI turns aerial drone video into instant, life-saving intelligence.*
>
> *With one click on our command map, our autonomous drone initiates a geofenced search sortie. Our integrated edge YOLOv8 model detects human silhouettes in real time, triggers acoustic warnings, and pins exact survivor coordinates on the map. With our instant 'Export Manifest' feature, ground rescue units receive precise, actionable waypoints immediately. RESQ-AI drastically cuts rescue response times and saves lives."*